Saarland University
Faculty of Mathematics and Computer Science
Department of Computer Science

Master's Thesis

Optimization-Based Precipitation Estimation from Commercial Microwave Links

submitted by

Iván Soto

Advisor

Prof. Dr.-Ing. Andreas Karrenbauer

Reviewers

Prof. Dr.-Ing. Andreas Karrenbauer (First Examiner)

Prof. Dr. Kurt Mehlhorn (Second Examiner)

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Abstract

Commercial microwave links provide an opportunistic source of rainfall information: rain attenuates the transmitted signal along each link path, and these attenuation measurements can be used to infer rainfall. Many existing CML rainfall-mapping approaches rely on link-level rain-rate estimates followed by spatial interpolation. In this thesis, we study optimization-based rainfall reconstruction from commercial microwave links as an alternative to purely interpolation-based map construction.

We construct a 100-patch benchmark from radar-derived European rainfall fields, simulate link attenuations using a physical attenuation model, and compare inverse-distance weighting (IDW), a link-aware interpolation variant (ILDW), and several optimization-based reconstructions. These include an ILDW-initialized nonlinear solver, a convex reformulation, a homotopy variant, and a ground-truth-initialized diagnostic solver. The evaluation compares pixelwise agreement with the radar-derived rainfall fields, distance-conditioned errors relative to link proximity, and agreement with the simulated link attenuations.

The results show that optimization-based reconstruction substantially improves upon IDW in patch-level RMSE and Pearson correlation. In particular, Convex Solver and Homotopy Solver reproduce the simulated link attenuations more accurately than both IDW and ILDW. At the same time, the case studies and attenuation-mismatch results show that minimizing the link-based objective is not always equivalent to recovering the radar-derived ground truth pixel by pixel. This highlights both the promise and the difficulty of formulating rainfall reconstruction from microwave-link measurements as an inverse optimization problem.

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I am very grateful to Guy Even for wholeheartedly bringing me on board for this project, putting lots of effort into convincing me that this was something exciting to work on, spending so much time with me along the way to push it forward, and sharing so much of his wisdom in every meeting we had.

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Chapter 1

Introduction

Rainfall is one of the most important variables in meteorology, yet measuring it accurately over a geographic area remains difficult. One approach that has been studied extensively is to use commercial microwave links as an indirect source of rainfall observations (Messer et al., 2006; Goldshtein et al., 2009; Chwala and Kunstmann, 2019; Lian et al., 2022). These links, originally installed for telecommunications, are attractive because they are already widespread, operate continuously, and provide measurements that carry information about rainfall without requiring a dedicated rainfall-sensor network. Because rain attenuates the transmitted signal along the link path, attenuation measurements collected on such links can be used to infer rainfall.

Reconstructing rainfall fields from microwave-link data has already been studied in the literature, often through interpolation-based methods that first convert each link into a representative rain-rate value and then spread those values across a region. Simple interpolation rules such as inverse-distance weighting (IDW) are attractive because they are straightforward to implement and provide a practical baseline for rainfall mapping (Lian et al., 2022).

In this thesis, we study whether rainfall reconstruction can be improved by replacing purely interpolative map construction with optimization-based reconstruction. Rather than treating the rainfall map only as an interpolation problem, we formulate reconstruction as an optimization problem that balances agreement with the observed link attenuations against terms that encourage spatially coherent rainfall fields. We evaluate several solver variants derived from this formulation and compare them against IDW, a standard interpolation baseline.

To test this question, we construct a 100-patch benchmark from radar-derived rainfall fields. We then simulate microwave-link attenuations with a physical model relating rainfall to link attenuation and compare IDW, ILDW, and several solver variants derived from the optimization-based formulation.

With this setup in place, the remainder of the thesis reviews the background and prior work, presents the benchmark and methods, reports the empirical comparison, and closes with a discussion of the main conclusions and limitations.

For reproducibility, the code, experiment configuration files, and generated report artifacts used for this thesis are available in the project repository at `https://github.com/guyeven/Rainfall-Estimation`.

Chapter 2

Background and Related Work

In most settings, rainfall is measured primarily with rain gauges and weather radar. Rain gauges provide accurate measurements at their installation sites, but gauge networks are often sparse across a region (McKee and Binns, 2016). Radar gives broad spatial coverage, but it does not measure rain rate directly and must infer it from electromagnetic echoes. This is one reason rainfall estimation remains challenging even where operational radar and gauge networks are already in place (Olsson et al., 2025).

Weather radar: strengths and limitations. A weather radar is a sensing system that monitors precipitation remotely by transmitting electromagnetic pulses and recording the energy that is scattered back toward the antenna. The return time gives distance, the antenna pointing gives direction, and the returned signal strength gives reflectivity. Rainfall is then inferred from that reflectivity through a conversion model. This is powerful because a single radar can observe precipitation over a wide region with high spatial and temporal resolution. At the same time, the inference is imperfect. The returned energy is influenced not only by rain, but also by the drop-size distribution and, in some situations, by snow, hail, insects, birds, terrain clutter, and other non-rain scatterers. Operational radar processing therefore includes filtering, classification, and calibration steps before reflectivity is turned into rainfall products.

Commercial microwave links as opportunistic sensors. A commercial microwave link is a fixed telecommunication path between two antennas, typically mounted on cell-phone towers. Rain along that path attenuates the transmitted microwave signal. By comparing the received signal during rainy periods with its dry-weather baseline, one can estimate the attenuation caused by rain along the link and then convert that attenuation into a path-averaged rain-rate estimate. This conversion relies on the standard power-law relationship between rain rate and microwave specific attenuation, which is introduced later in this work and itself goes back much earlier, including classical work such as Olsen et al. (1978). The specific idea of using measurements from wireless communication networks for rainfall monitoring is usually traced to Messer et al. (2006). Since then, the topic has been studied extensively and is now a well-established opportunistic-sensing approach for rainfall observation (Goldshtein et al., 2009; Chwala and Kunstmann, 2019; Olsson et al., 2025).

What information a link carries. In the present context, a link is modeled geometrically as the line segment connecting its two endpoints. Besides those endpoint coordinates, each link carries a carrier frequency and a polarization. The frequency determines the microwave band at which the signal is transmitted. The polarization describes the orientation of the electric field of the transmitted wave, typically horizontal or vertical. Both frequency and polarization matter because the ITU attenuation coefficients depend on them: the same rain rate leads to different specific attenuation at different frequencies and polarizations.

From link attenuation to rainfall maps. A microwave link does not provide a point measurement of rainfall. Instead, its attenuation reflects the combined effect of rain along the path between its two antennas, so different spatial rainfall patterns can produce similar link measurements. This ambiguity makes map reconstruction difficult. In practice, many approaches handle the problem with simple but scalable map-construction strategies. A common approach is to turn each link into a representative rainfall value and then interpolate those values across space, for example with Kriging or inverse-distance weighting (IDW) (Chwala and Kunstmann, 2019; Olsson et al., 2025). These methods are attractive because they are easy to implement, computationally light, and well suited to larger networks. We use IDW as the main interpolation baseline because it is a simple representative of this interpolation-based approach. We also evaluate ILDW, a link-aware variant that keeps the full link segment in the interpolation step; both methods are defined in the methodology section.

What has been done before. Early work first showed that CML signal measurements could be used to estimate rainfall over two-dimensional domains (Goldshtein et al., 2009). Later studies tested these ideas on larger link networks and in settings closer to operational rainfall monitoring. For example, Roversi et al. applied the RAINLINK workflow to 357 links in northern Italy and compared the resulting CML-derived rainfall maps with operational radar products from the regional weather service. They found that the CML-derived maps showed rainfall patterns broadly consistent with the radar products, while also providing an independent estimate of rainfall from the link network (Roversi et al., 2020). At a broader level, Olsson et al. survey opportunistic rainfall observations, including CML-based approaches, and emphasize both their scientific promise and the remaining challenges in data access, quality assurance, and operational integration (Olsson et al., 2025).

The literature establishes CMLs as useful opportunistic rain sensors, but it also leaves methodological choices in how link measurements are turned into maps. In this thesis, we use a controlled benchmark for CML-based rainfall mapping to ask whether optimization-based reconstruction can improve on interpolation-based map construction.

Chapter 3

Methodology

3.1 Data and Benchmark Construction

Our benchmark is built patch by patch. A *patch* is a rectangular crop from a one-hour rainfall accumulation map in the OPERA-based EURADCLIM radar rainfall dataset[1] (Overeem et al., 2025), with accumulation expressed in millimetres. Each patch is selected to capture a detected rain event, with its width and height constrained to lie between 50 km and 95 km. The 100 patches are extracted from 84 distinct hourly EURADCLIM maps: 70 maps contribute one patch, 12 maps contribute two patches, and 2 maps contribute three patches. Because the patch detector is designed to capture rainy events rather than dry background, the benchmark is rain-dominated: across the 100 patches, the mean non-rainy proportion is about 10.50%, where a pixel is classified as non-rainy when its ground-truth rain rate is below 0.6 mm/h.

For each benchmark patch, we extract the corresponding rainfall crop from the EURADCLIM map. We then place this field on a finer 125 m grid by dividing each 2 km pixel into 16×16 subpixels, each assigned its parent pixel's rainfall value. We smooth this field with a normalized Gaussian filter having a standard deviation of one refined pixel (125 m). The filter uses a 7×7-pixel neighborhood, extending 375 m in each grid direction, and nearest-value padding at the patch boundaries. It redistributes rainfall among neighboring pixels, although the boundary padding means that the total rainfall is not necessarily preserved exactly. This pragmatic smoothing step softens the artificial block boundaries created by the subdivision. The resulting refined rainfall field serves as the benchmark ground truth for simulating link attenuations and evaluating the solvers.

The microwave-link geometry is not geographically co-located with the selected EURADCLIM patch locations. Instead, we use records from the 4TU-NL dataset (Overeem, 2023), which describes an operational CML network covering the Netherlands and provides the endpoint locations and operating frequencies of its links. We use these endpoint locations and frequencies as a realistic network template, place this network in the coordinate system of each selected patch, and retain only links whose endpoints fall inside the patch. Thus, the link geometries and frequencies are derived from actual Dutch CML infrastructure, whereas their placement relative to the EURADCLIM rainfall fields is synthetic. Across the 100 patches, this procedure retains between 939 and 1996 links per patch (mean 1324). The attenuation of each retained link is then simu-

[1] OPERA (Operational Programme for the Exchange of Weather Radar Information) is a programme of EUMETNET, a network of European national meteorological and hydrological services, that coordinates the exchange and compositing of weather-radar observations across Europe. EURADCLIM (European Climatological Gauge-Adjusted Radar Precipitation Dataset) is a processed historical precipitation dataset derived from OPERA radar composites, which are further cleaned and adjusted using rain-gauge observations.

lated from the radar-derived rainfall field rather than taken from the original operational CML measurements.

The 4TU-derived link file used in this benchmark does not specify polarization, so the patch-export pipeline supplies a default polarization when missing. In the present benchmark, that default is set to horizontal (H), and the forward model therefore uses the ITU coefficients corresponding to horizontal polarization.

To give a concrete sense of the benchmark, we next characterize the selected patches by their geographic distribution, dimensions, and retained 4TU-derived link counts.

3.1.1 Geographic Distribution of the Benchmark Patches

Figure 1 shows that the selected rainy-event patches are not concentrated in one country or local region; instead, they are spread across northern, western, central, southern, and southeastern Europe.

Figure 1: Geographic distribution of the 100 benchmark patches over the OPERA domain. Each red rectangle marks the footprint of one selected patch.

3.1.2 Patch-Geometry Overview

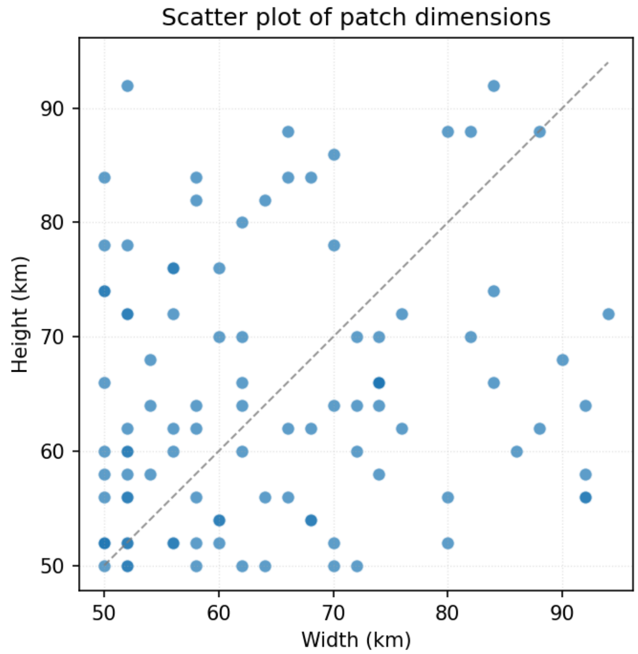


Figure 2: Patch width versus patch height for the 100 benchmark patches, with one dot per patch. Across patches, the mean patch area is 269,491 pixels with SD of 76,755 pixels (pixels are squares with side length 125 m).

Patch size also determines how many overlaid 4TU-derived links are retained, since only links with both endpoints inside the patch are used. Some retained links appear at more than one frequency, so Table 1 reports representative link counts and one- versus two-frequency counts for three patch sizes.

Patch size (km)	# links	1-multiplicity	2-multiplicity
50×50	939	99	420
62×60	1254	128	563
72×70	1507	157	675

Table 1: Representative link multiplicities for three patch dimensions. The multiplicity of a link is defined as the number of frequencies assigned to it. The second column reports the total number of link–frequency pairs. In the data set, link multiplicities were either 1 or 2, with multiplicity-one links accounting for approximately 10% of the reported link–frequency pairs.

3.2 Forward Model

In the benchmark, link attenuations are generated from the known benchmark ground-truth rainfall field using the forward model. Each link is represented as a line segment between its endpoints, and its attenuation is computed using ITU-R P.838-3, the standard relation for rain-induced specific attenuation in microwave links (International Telecommunication Union, 2005). In this model, the specific attenuation (dB/km) is

$$\gamma_R = \kappa(f, \mathrm{pol})\, R^{\alpha(f,\mathrm{pol})},$$

where R is rain rate (mm/h), and (κ, α) depend on link frequency and polarization. For this calculation, each one-hour accumulation value is treated as the corresponding hourly mean rain rate. The total link attenuation is then the line integral of specific attenuation along the propagation path,

$$A_\ell \triangleq \int_\ell \gamma_R\big(R(s)\big)\, \mathrm{d}s.$$

In the benchmark, and likewise for the methods we compare against, this integral is evaluated in discretized form on the refined 125 m grid: each link is traced through crossed grid cells, and attenuation is accumulated as a length-weighted sum of cellwise specific attenuations,

$$A_\ell \approx \sum_{c \in \mathcal{C}(\ell)} \kappa_\ell\, R_c^{\alpha_\ell}\, \Delta s_{\ell,c},$$

where $\mathcal{C}(\ell)$ is the set of crossed cells and $\Delta s_{\ell,c}$ is the path length of link ℓ inside cell c (in km). These computed attenuation values, together with each link's endpoint coordinates, frequency, and polarization, form the inputs to the inverse rainfall reconstruction.

3.3 Inverse Solvers

After the forward model defines how a rainfall field produces link attenuations, the reconstruction task runs in the opposite direction: given link-level observations, estimate the rainfall field that produced them. We compare IDW, ILDW, and four optimization-based reconstructions for this inverse problem: Solver(ILDW), Convex Solver, Homotopy Solver, and Solver(GT). Solver(GT) is initialized from the ground-truth rainfall field, so it is not a method that could be used on real unknown rainfall fields. Instead, we use it as an optimistic diagnostic reference for this objective and optimizer.

Using the same link-level observations and forward-model conventions described above, we now describe each solver family and its implementation details.

3.3.1 IDW

Inverse-distance weighting (IDW) is one of the standard simple interpolation strategies used in commercial-microwave-link rainfall reconstruction (Chwala and Kunstmann, 2019). The input consists of a set of links, where each link carries total attenuation, frequency, and polarization.

IDW first converts each observed link attenuation A_ℓ into a single link-level rain-rate estimate defined as:

$$R_\ell \triangleq \left(\frac{A_\ell/|\ell|}{\kappa_\ell}\right)^{1/\alpha_\ell}.$$

where A_ℓ denotes the attenuation of link ℓ, $|\ell|$ its length, and $(\kappa_\ell, \alpha_\ell)$ the ITU-R P.838-3 coefficients corresponding to its frequency and polarization.

Each per-link rain-rate estimate is then treated as a virtual point observation at the link midpoint. For each pixel p with center c_p, IDW computes a weighted average of nearby R_ℓ values, using weights proportional to $\|m_\ell - c_p\|_2^{-2}$ within a prescribed search radius $r_{\max}$ (set to 15 km in our study). If one or more link midpoints lie inside pixel p, $\hat{R}_p$ (the estimated rainfall in pixel p), is set to the mean of their rainfall estimates. This defines the discretized IDW baseline used throughout this thesis.

Algorithm 1: Discretized IDW. Each link is replaced with a virtual gauge located at the midpoint of the link. The rain rate at the gauge is set to the uniform rain rate that reproduces the input link attenuation under ITU-R P.838-3. IDW is then applied with respect to these virtual gauges. We set the exponent of the decay to 2.

Data: link set $\mathcal{L}$, a patch P, and a maximum influence distance $r_{\max}$
Result: rainfall field $\hat{R}$

foreach *link* $\ell \in \mathcal{L}$ **do** `// compute rain rate per link`
- $(\ell.\kappa, \ell.\alpha) \leftarrow \mathrm{ITU838}(\ell.\mathrm{freq}, \ell.\mathrm{polarization})$
- $\ell.\mathrm{rainrate} \leftarrow \left(\dfrac{\ell.\mathrm{attenuation}/\ell.\mathrm{length}}{\ell.\kappa}\right)^{1/\ell.\alpha}$

foreach *pixel* $p \in P$ **do** `// compute rain` $\hat{R}_p$
- $M_p \leftarrow \{\ell \in \mathcal{L} : \ell.\mathrm{midpoint} \in p\}$
- **if** $M_p \neq \emptyset$ **then**
  - $\hat{R}_p \leftarrow \dfrac{1}{|M_p|} \sum_{\ell \in M_p} \ell.\mathrm{rainrate}$ `// pixel contains a link midpoint`
- **else**
  - $N_p \leftarrow \{\ell \in \mathcal{L} : \|\ell.\mathrm{midpoint} - p.\mathrm{center}\|_2 \leq r_{\max}\}$
  - **if** $N_p = \emptyset$ **then**
    - $\hat{R}_p \leftarrow 0$ `// pixel far from links`
  - **else**
    - $\hat{R}_p \leftarrow \sum_{\ell \in N_p} \dfrac{\|\ell.\mathrm{midpoint} - p.\mathrm{center}\|_2^{-2}}{\sum_{\ell' \in N_p} \|\ell'.\mathrm{midpoint} - p.\mathrm{center}\|_2^{-2}} \cdot \ell.\mathrm{rainrate}$

3.3.2 ILDW

ILDW is a link-aware modification of IDW used in this work. It starts from the same per-link rain-rate values obtained by inverting the discretized attenuation forward model for each link. The motivation for this variant is that a microwave-link attenuation measurement is associated with an entire path, not only with the midpoint of that path. We therefore considered whether interpolation could be improved by using the full link geometry when deciding which links should influence a pixel and how strongly they should do so.

More precisely, for a pixel center c_p, ILDW replaces the midpoint distance $\|m_\ell - c_p\|_2$ by the point-to-segment distance

$$\mathrm{dist}(c_p, \ell),$$

where $\ell = [p_{\ell,1}, p_{\ell,2}]$ denotes the full link segment with endpoint coordinates $p_{\ell,1}$ and $p_{\ell,2}$.

If one or more links cross the pixel, the estimate is taken to be the mean of the corresponding link rain-rate values. Otherwise, the estimate is a weighted average over nearby links within radius $r_{\max}$ (set to 15 km), with weights proportional to the inverse squared segment distance. We considered this variant important because it preserves the same simple interpolation philosophy as IDW while using more of the link-path geometry.

At the algorithmic level, ILDW differs from Algorithm 1 only in how link geometry enters the interpolation step: midpoint membership and midpoint distance are replaced by pixel–segment intersection and distance to the full link segment.

Algorithm 2: Discretized ILDW. Compared with Algorithm 1, the red lines replace midpoint-based interpolation by interpolation based on the full link segment.

Data: link set $\mathcal{L}$, a patch P, and a maximum influence distance $r_{\max}$
Result: rainfall field $\hat{R}$

foreach *link* $\ell \in \mathcal{L}$ **do** `// compute rain rate per link`
  $(\ell.\kappa, \ell.\alpha) \leftarrow \text{ITU838}(\ell.\text{freq}, \ell.\text{polarization})$
  $\ell.\text{rainrate} \leftarrow \left(\dfrac{\ell.\text{attenuation}/\ell.\text{length}}{\ell.\kappa}\right)^{1/\ell.\alpha}$
foreach *pixel* $p \in P$ **do** `// compute rain` $\hat{R}_p$
  $X_p \leftarrow \{\ell \in \mathcal{L} : \ell \cap p \neq \emptyset\}$
  if $X_p \neq \emptyset$ **then**
    $\hat{R}_p \leftarrow \dfrac{1}{|X_p|} \sum_{\ell \in X_p} \ell.\text{rainrate}$ `// at least one link traverses the pixel`
  else
    $N_p \leftarrow \{\ell \in \mathcal{L} : \text{dist}(p, \ell) \leq r_{\max}\}$
    if $N_p = \emptyset$ **then**
      $\hat{R}_p \leftarrow 0$ `// pixel far from links`
    else
      $\hat{R}_p \leftarrow \sum_{\ell \in N_p} \dfrac{\text{dist}^{-2}(p.\text{center}, \ell)}{\sum_{\ell' \in N_p} \text{dist}^{-2}(p.\text{center}, \ell')} \cdot \ell.\text{rainrate}$

3.3.3 Optimization-Based Reconstruction

The optimization-based reconstructions explore a different approach from purely interpolation-based rainfall mapping. IDW and ILDW first reduce each link attenuation to a single link-level rain-rate estimate and then spread that information spatially by geometric weighting. In contrast, the optimization-based methods treat the rainfall value in every pixel as an unknown and choose the full rainfall field by minimizing an objective that combines agreement with the observed link attenuations with terms that encourage spatially coherent solutions.

Optimization Objective For each patch P, we identify P with its set of pixels, and denote by $\mathcal{L}(P)$ the set of links contained in P. Let $r \in \mathbb{R}_{\geq 0}^{|P|}$ denote the unknown rainfall field written as a vector over pixels. We solve the constrained optimization problem

$$\min_{r \in \mathbb{R}_{\geq 0}^{|P|}} J(r),$$

where the objective is a weighted sum of four terms:

$$J(r) \triangleq \alpha_{\text{atten}} J_{\text{atten}}(r) \; + \; \alpha_{1d} J_{1d}(r) \; + \; \alpha_{2d} J_{2d}(r) \; + \; \alpha_{\text{total}} J_{\text{total}}(r).$$

Here J_{atten} penalizes disagreement between observed and forward-model-predicted link attenuations, while J_{1d}, J_{2d}, and J_{total} are regularization terms that encourage the estimated rainfall field to remain spatially smooth and free of unnecessary oscillation, while weakly favoring smaller overall rainfall magnitudes. In the normalized solver studied here, we define the weights instance by instance using the corresponding objective values at the ILDW solution:

$$\alpha_{\text{atten}} \triangleq \frac{1}{J_{\text{atten}}(\text{ILDW})}, \qquad \alpha_{1d} \triangleq \frac{1}{J_{1d}(\text{ILDW})},$$
$$\alpha_{2d} \triangleq \frac{1}{2} \cdot \frac{1}{J_{2d}(\text{ILDW})}, \quad \alpha_{\text{total}} \triangleq \frac{1}{|P|} \cdot \frac{1}{J_{\text{total}}(\text{ILDW})}.$$

More concretely, J_{atten} is the attenuation-misfit term:

$$J_{\text{atten}}(r) \triangleq \frac{1}{|\mathcal{L}(P)|} \sum_{\ell \in \mathcal{L}(P)} \frac{\left(\widehat{A}_\ell(r) - A_\ell^{\text{obs}}\right)^2}{|\ell|},$$

where $\widehat{A}_\ell(r)$ is the attenuation predicted by the forward model from the rainfall field r, A_ℓ^{obs} is the observed attenuation of link ℓ, and $|\ell|$ is the link length in km. Thus J_{atten} measures how far the forward-model-implied attenuations are from the observed attenuations, with a normalization by link length and averaging over the links in the patch.

In this benchmark, A_ℓ^{obs} is generated from the benchmark ground truth, but, except for the diagnostic Solver(GT), the reconstruction methods receive only these attenuation values and the corresponding link information. In an operational setting, A_ℓ^{obs} would instead be estimated from actual CML signal measurements after dry-baseline estimation and correction or filtering of non-rain effects, so the attenuation-misfit term would not require radar data as an input.

The term J_{1d} is a first-difference penalty:

$$J_{1d}(r) \triangleq \frac{1}{|P|} \sum_{(u,v) \in \mathcal{E}(P)} \left(r_u - r_v\right)^2,$$

where $\mathcal{E}(P)$ is the set of horizontally and vertically adjacent pixel pairs in patch P. This term sums squared differences between neighboring pixel values and therefore discourages abrupt pixel-to-pixel jumps in the estimated rainfall field.

The term J_{2d} is a second-difference penalty:

$$J_{2d}(r) \triangleq \frac{1}{|P|} \sum_{(p,q,t) \in \mathcal{T}(P)} \left((r_q - r_p) - (r_t - r_q)\right)^2,$$

where $\mathcal{T}(P)$ is the set of horizontal and vertical triples of collinear pixels in patch P. It penalizes changes in the first difference, so it discourages unnecessary curvature or oscillation along rows and columns.

Finally, J_{total} is an ℓ_2 shrinkage term on the rainfall field itself:

$$J_{\text{total}}(r) \triangleq \frac{1}{|P|} \sum_{p \in P} r_p^2.$$

This term biases the solution toward smaller overall rainfall values, except where larger values are needed to match the observed link attenuations.

These four terms are added together only after weighting, so the optimizer balances attenuation fit against smoothness, curvature control, and shrinkage.

These weights should be read as heuristic normalization choices rather than as parameters derived from an optimality principle. Using the ILDW objective values puts the four terms on comparable numerical scales at the same reference field, so that the optimizer is not dominated by one raw term simply because of units or magnitude. The additional factor 1/2 downweights the second-difference penalty relative to the first-difference penalty, while the factor $1/|P|$ makes the global shrinkage term a weak regularizer. We did not perform a systematic sensitivity analysis over nearby weight choices.

Solver Variants All optimization-based variants are solved iteratively with L-BFGS-B, a limited-memory quasi-Newton method for smooth constrained optimization (Byrd et al., 1995; Zhu et al., 1997). Starting from an initial rainfall field, the optimizer repeatedly updates the field to reduce the objective while enforcing the nonnegativity bounds on rainfall. In the present setting, the optimization variable is the rainfall value at every pixel of the patch. We write

$$\hat{r} \leftarrow \text{L-BFGS-B}(\mathcal{L}, P, r_{\text{init}})$$

to denote one invocation of L-BFGS-B on patch P with link set $\mathcal{L}$ and initial rainfall field r_{init}.

Solver(ILDW) The first variant applies L-BFGS-B directly to the original nonconvex objective above, using the ILDW rainfall field as the initial point:

$$\hat{r}_{\text{ILDW}} \leftarrow \text{L-BFGS-B}(\mathcal{L}, P, r_{\text{ILDW}}).$$

The objective is generally nonconvex because the ITU-R attenuation model uses a power law in rainfall. For a link ℓ, the predicted attenuation has the form

$$\widehat{A}_\ell(r) = \sum_{c \in \mathcal{C}(\ell)} \kappa_\ell r_c^{\alpha_\ell} \Delta s_{\ell,c}.$$

When $\alpha_\ell = 1$, this predicted attenuation is linear in the pixel rain rates, so the squared attenuation residual is convex. For $\alpha_\ell \neq 1$, the attenuation residual is nonlinear and the resulting multivariable squared-residual term is not generally convex. As a simple scalar example, when $\alpha > 1$ and $A > 0$, $(r^\alpha - A)^2$ has negative curvature for sufficiently small positive r. Thus the solution found by L-BFGS-B can depend on the initial field.

We use ILDW as the initial point for this nonconvex solve because it is already constructed to be consistent with individual link attenuations in the simple case where links do not overlap. If links were pairwise disjoint at the pixel level, assigning each crossed pixel the rain-rate estimate of the link that crosses it would reproduce each observed attenuation exactly under the same forward model. Real patches contain overlapping links, so this exact consistency cannot generally hold for all links at once, but ILDW remains a natural initialization because it is organized around the same link paths whose attenuations the optimizer tries to fit.

Convex Solver The second variant constructs a convex surrogate inverse problem by replacing each original link with a virtual link. The virtual link keeps the same geometry and length, but uses a reference frequency f^0 for which the ITU-R attenuation exponent is one. In the present benchmark the links are treated as horizontally polarized, for which this reference frequency is $f^0 = 24.92\,\text{GHz}$. For each original link, we first compute the uniform rain-rate equivalent

$$R_\ell \triangleq \left(\frac{A_\ell^{\text{obs}}}{|\ell|\,\kappa_\ell} \right)^{1/\alpha_\ell},$$

that is, the constant rain rate along the link that would reproduce its observed attenuation under the original link coefficients. The virtual observed attenuation is then

$$A_\ell^0 \triangleq |\ell|\,\kappa_0 R_\ell,$$

where κ_0 is the ITU-R coefficient at f^0. We denote the resulting virtual link set by $\mathcal{L}^0$. Because the attenuation exponent is one for these virtual links, the predicted attenuation is linear in the rainfall field, so the attenuation-misfit term is a sum of squared linear residuals. Together with the convex regularization terms, this gives a convex surrogate objective. We write

$$\hat{r}^0 \leftarrow \text{L-BFGS-B}(\mathcal{L}^0, P, r_{\text{init}})$$

for this solve. In the final experiments, r_{init} is the constant field 0.6 mm/h. This is the method reported as Convex Solver.

For Convex Solver, the attenuation weight is normalized within the virtual-link formulation:

$$\alpha_{\text{atten}}^{\text{Convex}} \triangleq \frac{1}{J_{\text{atten}}^{\text{virtual}}(r_{\text{ILDW}})},$$

where $J_{\text{atten}}^{\text{virtual}}$ is evaluated using the virtual link set $\mathcal{L}^0$ and virtual observed attenuations A_ℓ^0. Thus, the ILDW rainfall field remains the reference field for normalization, but the denominator is its attenuation mismatch under the virtual-link model rather than $J_{\text{atten}}(r_{\text{ILDW}})$ for the original links.

Homotopy Solver The third variant uses homotopy continuation (Allgower and Georg, 2003) to move gradually from the convex surrogate problem back toward the original nonconvex formulation. For $\lambda \in [0, 1]$, define a virtual link set $\mathcal{L}^\lambda$ in which each link keeps the same geometry, length, and polarization as the original link, but has interpolated frequency

$$f_\ell^\lambda \triangleq (1-\lambda) f^0 + \lambda f_\ell.$$

Let $\kappa_\ell(\lambda)$ and $\alpha_\ell(\lambda)$ be the ITU-R coefficients at frequency f_ℓ^λ. The corresponding virtual observed attenuation is

$$A_\ell^\lambda \triangleq |\ell|\,\kappa_\ell(\lambda) R_\ell^{\alpha_\ell(\lambda)}.$$

Starting from the convex problem at $\lambda = 0$, the solver advances through a sequence of increasing λ values, each time initializing from the previous solution:

$$\hat{r}^0 \leftarrow \text{L-BFGS-B}(\mathcal{L}^0, P, r_{\text{init}}), \qquad \hat{r}^\lambda \leftarrow \text{L-BFGS-B}(\mathcal{L}^\lambda, P, \hat{r}^{\lambda-\delta}).$$

The final output is $\hat{r}^1$, which corresponds to the original link frequencies. The first stage uses the same constant 0.6 mm/h initial field as Convex Solver. At each subsequent stage, the previous stage's solution provides only the initial field. The attenuation term and its normalization correspond to the transformed link model at the current value of λ:

$$\alpha_{\text{atten}}^\lambda \triangleq \frac{1}{J_{\text{atten}}^\lambda(r_{\text{ILDW}})}.$$

Thus, the same ILDW rainfall field serves as the normalization reference at every stage, while J_{atten}^λ is evaluated using that stage's link model and virtual observed attenuations. In the final experiments, this continuation-based method is reported as Homotopy Solver, and we use a continuation step size of $\delta = 0.1$.

Solver(GT) We also include a ground-truth-initialized diagnostic run:

$$\hat{r}_{\mathrm{GT}} \leftarrow \text{L-BFGS-B}(\mathcal{L}, P, r_{\mathrm{GT}}).$$

This run uses information unavailable in practice, so it is not a deployable reconstruction method. Instead, it provides an optimistic empirical reference for interpreting the objective landscape and for seeing what the objective preserves, or moves away from, when initialized at the target rainfall field.

Chapter 4

Evaluation

4.1 Evaluation Protocol

We evaluate each solver in four ways: overall agreement with the ground-truth rainfall map, how errors change as distance from links increases, wet/dry classification behavior, and agreement with observed link attenuations under the forward model. We first report patch-level summaries across all 100 benchmark patches, and then provide qualitative case studies in the appendix. We additionally report end-to-end runtime for IDW, ILDW, and Convex Solver to compare the time required by each method to reconstruct a patch. All optimization-based results reported below use the objective terms and normalization rules described above. The single-stage variants have a maximum budget of 2000 L-BFGS-B iterations per patch, while the Homotopy Solver applies this limit separately at each continuation stage.

4.1.1 Optimization Stopping Criteria

The 2000-iteration budget was used as a cap rather than as a fixed number of iterations. Every L-BFGS-B invocation, including each stage of the Homotopy Solver, used the same stopping tolerances, with relative objective-reduction tolerance $\mathrm{ftol} = 10^{-6}$ and projected-gradient tolerance $\mathrm{gtol} = 10^{-6}$. Let J_k denote the objective value at iteration k. The ftol criterion stops the solve when the relative objective decrease between two successive iterations is small,

$$\frac{J_k - J_{k+1}}{\max\{|J_k|, |J_{k+1}|, 1\}} \leq \mathrm{ftol}.$$

The gtol criterion stops the solve when the projected gradient is small,

$$\|g_{k+1}^{\mathrm{proj}}\|_\infty \leq \mathrm{gtol},$$

where g_{k+1}^{proj} is the gradient projected according to the bound constraints used by L-BFGS-B. The ftol criterion establishes that the relative objective reduction has become small, whereas satisfying gtol provides the stronger indication that the bound-constrained optimizer is close to a stationary point. No Solver(ILDW), Convex Solver, or Solver(GT) reconstruction reached its 2000-iteration cap, and no Homotopy Solver stage reached the corresponding per-stage cap. Thus, none of the reported reconstructions terminated because it exhausted the configured iteration budget. Because nearly all runs stopped through ftol, these stop reasons indicate negligible objective change between successive iterations, but not necessarily a small projected gradient.

Solver	Mean iter.	SD iter.	Min	Max	Final stop reason
Solver(ILDW)	557.4	289.1	3	1198	100 ftol
Convex Solver	936.7	233.8	387	1557	99 ftol, 1 gtol
Homotopy Solver	1074.6	243.0	430	1715	100 ftol
Solver(GT)	594.4	345.9	3	1787	100 ftol

Table 2: Iteration counts for the optimization-based reconstruction runs across the 100 evaluated patches. For the single-stage variants, the count is the number of iterations in that patch's L-BFGS-B invocation. For Homotopy Solver, it is the total number of inner L-BFGS-B iterations summed across all continuation stages, while the displayed stop reason refers to the final stage. The standard deviation is computed across patches. No single-stage reconstruction or Homotopy stage reached its applicable 2000-iteration cap.

4.2 Patch-Level Metrics

We now turn to the empirical comparison on the hundred-patch benchmark. We begin with whole-patch summaries, then examine how pixel-level errors vary with link proximity, and finally report attenuation-consistency analyses.

For a patch P, let $R_P(p)$ denote the ground-truth rainfall at pixel p, and let $\widehat{R}_{P,s}(p)$ denote the rainfall predicted at pixel p by solver s. We compute three scalar metrics over all pixels in P: the root mean squared error

$$\mathrm{RMSE}_{P,s} \triangleq \sqrt{\frac{1}{|P|}\sum_{p\in P}\big(\widehat{R}_{P,s}(p) - R_P(p)\big)^2},$$

the signed bias

$$\mathrm{Bias}_{P,s} \triangleq \frac{1}{|P|}\sum_{p\in P}\big(\widehat{R}_{P,s}(p) - R_P(p)\big),$$

and the Pearson correlation coefficient

$$r_{P,s} \triangleq \frac{\sum_{p\in P}\big(\widehat{R}_{P,s}(p) - \overline{\widehat{R}}_{P,s}\big)\big(R_P(p) - \overline{R}_P\big)}{\sqrt{\sum_{p\in P}\big(\widehat{R}_{P,s}(p) - \overline{\widehat{R}}_{P,s}\big)^2}\,\sqrt{\sum_{p\in P}\big(R_P(p) - \overline{R}_P\big)^2}},$$

where

$$\overline{\widehat{R}}_{P,s} \triangleq \frac{1}{|P|}\sum_{p\in P}\widehat{R}_{P,s}(p), \qquad \overline{R}_P \triangleq \frac{1}{|P|}\sum_{p\in P}R_P(p).$$

RMSE measures overall magnitude error. Bias measures average signed error (net over- or underestimation). Correlation measures how well the spatial structure of the field is reproduced independently of scale and offset. The Pearson correlation coefficient satisfies $r_{P,s} \in [-1, 1]$. A value of 1 indicates perfect positive linear agreement between predicted and ground-truth rainfall maps. A value of 0 indicates no linear relationship. A value of -1 indicates perfect negative linear agreement (an exact inverse pattern).

Solver	RMSE (mm/h)	Signed bias (mm/h)	Correlation
IDW	1.307 ± 1.234	0.034 ± 0.296	0.866 ± 0.059
ILDW	1.239 ± 1.182	0.047 ± 0.261	0.878 ± 0.055
Solver(ILDW)	1.059 ± 1.192	0.040 ± 0.235	0.918 ± 0.055
Convex Solver	0.845 ± 0.803	0.023 ± 0.075	0.940 ± 0.034
Homotopy Solver	0.845 ± 0.803	0.023 ± 0.075	0.940 ± 0.034
Solver(GT)	0.503 ± 0.369	0.009 ± 0.030	0.961 ± 0.032

Table 3: Descriptive patch-level summary of three global map metrics across the 100 evaluated patches. Each entry reports mean ± standard deviation across patches. Here RMSE and signed bias are computed in the same rainfall units as R_P and $\widehat{R}_{P,s}$, while Correlation is the Pearson correlation coefficient between the predicted and ground-truth rainfall fields within each patch.

Table 3 gives a descriptive overview of the scale and variability of the patch-level metrics for each solver. Homotopy Solver gives nearly identical descriptive metrics to Convex Solver, so in these experiments its added continuation machinery does not produce an additional empirical gain in the map-level summaries. Solver(GT) has the lowest mean RMSE, lowest mean signed bias, and highest mean correlation, as expected for a run initialized from the ground truth. Comparing IDW and ILDW also suggests that keeping more of the link-path geometry can be useful even within a simple interpolation framework: ILDW has lower mean RMSE and higher mean correlation than midpoint IDW, although its mean signed bias is not lower than IDW's.

Because all solvers are evaluated on the same 100 patches, direct solver comparisons can use paired patchwise differences. We treat these differences as approximately independent because the patches are distinct spatial crops and are often widely separated, even when they come from the same hourly EURADCLIM map. Among the 18 pairs of patches extracted from the same hourly EURADCLIM map, the median distance between patch centers is approximately 544 km, and only two pairs overlap spatially. The independence assumption may not hold exactly, however. Considering patches extracted from different hourly maps, 16 pairs both share at least one native EURADCLIM grid cell and occur within six hours of each other; six of these pairs are separated by exactly one hour. These pairs could represent successive observations of the same evolving rainfall system, and their solver differences could therefore be correlated.

We restrict the paired comparison to Convex Solver against IDW, since this is the main optimized solver relative to the standard interpolation baseline. Convex Solver has lower RMSE and higher Pearson correlation on all 100 patches. For absolute bias, Convex Solver is closer to zero on 78 patches and IDW is closer on 22 patches.

Unlike the RMSE and correlation comparisons, the absolute-bias comparison does not favor Convex Solver for every patch. We therefore use a one-sided paired t-test, implemented as a one-sample t-test on the patch-level paired differences, to assess whether Convex Solver has lower absolute bias than IDW on average rather than interpreting the win counts alone. Because this test assumes that the patch-level paired differences are independent, its p-value should be interpreted cautiously in light of the possible spatiotemporal dependence noted above.

For absolute bias, we define

$$d_P = |\mathrm{Bias}_{P,\mathrm{Convex}}| - |\mathrm{Bias}_{P,\mathrm{IDW}}|,$$

where a negative value means that Convex Solver is closer to zero bias on patch P. We test $H_0 : \mathbb{E}[d_P] \geq 0$ against $H_1 : \mathbb{E}[d_P] < 0$ using the 100 patch-level differences. The test rejects the null hypothesis at $\alpha = 0.01$ ($p = 4.667 \times 10^{-7}$).

Metric	Better direction	Convex wins	IDW wins	Patch-level evidence
RMSE	lower	100	0	all patches favor Convex Solver
Absolute Bias	lower	78	22	rejects null: $p = 4.667 \times 10^{-7}$
Pearson correlation	higher	100	0	all patches favor Convex Solver

Table 4: Paired patchwise comparison between Convex Solver and IDW across the 100 evaluated patches.

4.2.1 Attenuation Mismatch

The previous metrics compare each reconstructed rainfall map directly to the ground-truth radar field. We also evaluate the reconstructions from the point of view of the CML observations themselves: if a predicted rainfall field is passed through the forward attenuation model, how closely does it reproduce the observed link attenuations? This matters because a useful reconstruction should not only resemble the radar field pixelwise, but should also remain consistent with the link observations that drive the inverse problem.

We report two complementary attenuation-mismatch summaries in the same table. The mean absolute attenuation error per kilometer gives a directly interpretable link-level discrepancy in dB/km, while J_{atten} gives a squared attenuation-mismatch measure. For every solver, both measures are evaluated using the original link frequencies, polarizations, and observed attenuations. Thus, J_{atten} is a shared post-hoc evaluation measure rather than necessarily the attenuation term optimized internally by a given solver. In particular, Convex Solver optimizes $J_{\text{atten}}^{\text{virtual}}$, while Homotopy Solver successively optimizes $J_{\text{atten}}^{\lambda}$ for multiple values of λ; the final reconstructions from both methods are evaluated here using J_{atten}.

Solver	Abs. err. mean (dB/km)	Abs. err. SD (dB/km)	J_{atten} mean (dB^2/km)	J_{atten} SD (dB^2/km)
IDW	0.05034	0.03396	0.05284	0.13575
ILDW	0.01356	0.00943	0.004204	0.01390
Solver(ILDW)	0.00207	0.00373	0.0002951	0.001466
Convex Solver	0.00107	0.00099	0.00003354	0.0001431
Homotopy Solver	0.00074	0.00067	0.00001332	0.00005244
Solver(GT)	0.00129	0.00166	0.00004951	0.0001492

Table 5: Patch-level attenuation-mismatch summaries across the 100 evaluated patches. For a given patch and solver s, the absolute-error columns summarize $\frac{1}{N_{\text{links}}} \sum_{\ell \in \text{links}} \frac{|\hat{A}_{\ell,s} - A_\ell^{\text{obs}}|}{|\ell|}$, while $J_{\text{atten}} = \frac{1}{N_{\text{links}}} \sum_{\ell \in \text{links}} \frac{\left(\hat{A}_{\ell,s} - A_\ell^{\text{obs}}\right)^2}{|\ell|}$. Here $\hat{A}_{\ell,s}$ is recomputed from the rainfall field estimated by solver s using the original link frequencies and polarizations, A_ℓ^{obs} is the observed attenuation, and $|\ell|$ is the link length in kilometers. Lower values indicate better agreement with the observed link attenuations.

Together, these attenuation-based summaries show that the optimized solvers match the observed link attenuations much more closely than IDW and ILDW, with Convex Solver and Homotopy Solver giving the smallest mismatches. They also add a useful qualification to the map-level results above: pixelwise agreement with the radar-derived ground truth and consistency with the simulated CML observations are two measures that capture different aspects of reconstruction quality. Solver(GT) remains closest to the radar field, but Convex Solver and Homotopy Solver better satisfy the link observations under the forward model. Thus, the field closest to the radar ground truth is not necessarily the field that minimizes attenuation mismatch. In addition, although Homotopy Solver does not improve the RMSE, bias, or correlation summaries relative to Convex Solver, it does provide a small gain in link consistency.

4.3 Distance-Conditioned Error Analysis

We next ask how prediction quality varies with link proximity. To answer this question, we group pixels into distance bins according to their distance to the third-closest link and summarize patch-level error statistics within each bin. We use distance to the third-closest link as a proxy for local link density: small values indicate several nearby links, whereas large values indicate sparse coverage. Unlike nearest-link distance, this measure does not classify a location as densely covered solely because one link is nearby. The choice of three is a practical analysis choice rather than a value dictated by the physical model.

We examine rainy pixels using relative absolute error and non-rainy pixels using absolute error. In this subsection we focus exclusively on distance-conditioned pixelwise error summaries.

4.3.1 Rainy Pixel Relative Absolute Error by Link Distance

We define a pixel p to be rainy if $R_P(p) \geq 0.6$ mm/h, and we write

$$P_{\text{rain}} = \{p \in P : R_P(p) \geq 0.6\}.$$

We use 0.6 mm/h as a pragmatic analysis threshold to exclude trace and very weak rainfall, for which relative errors are highly sensitive to small absolute deviations and operational CML detection becomes less reliable (Polz et al., 2020). The threshold does not represent a universal physical boundary between rain and no rain, so the wet/dry summaries below should still be interpreted as threshold-dependent.

For a rainy pixel p, the relative absolute error with respect to solver s is

$$\mathrm{RAE}_{P,s}(p) \triangleq \frac{|R_P(p) - \widehat{R}_{P,s}(p)|}{R_P(p)}.$$

To study how prediction quality changes with link proximity, we group pixels by distance to the third-closest link. For a pixel $p \in P$, let $c(p) \in \mathbb{R}^2$ denote the center of pixel p. For a link $\ell \in \mathcal{L}(P)$, viewed geometrically as a line segment, let

$$\mathrm{dist}(p, \ell) \triangleq \mathrm{dist}\big(c(p), \ell\big)$$

denote the Euclidean distance from the pixel center to that link segment. If the links in patch P are written as $\ell_1, \ldots, \ell_{|\mathcal{L}(P)|}$, we define

$$d_3(p) \triangleq \text{the third-order statistic of } \big\{\mathrm{dist}(p, \ell_j)\big\}_{j=1}^{|\mathcal{L}(P)|},$$

that is, the distance from the center of pixel p to its third-closest link segment.[2]

For each index i, let b_{i-1} and b_i denote the lower and upper distance cutoffs of the ith bin, and write $B_i = (b_{i-1}, b_i]$. A pixel p belongs to bin B_i exactly when $d_3(p) \in B_i$.

[2] For computational efficiency, $d_3(p)$ is approximated using a sampled candidate search, followed by exact point-to-segment distance calculations for the candidate links.

For each patch P, solver s, and distance bin B_i, we then collect the rainy-pixel RAE values in that bin and summarize them by the patch-level median

$$m_{s,i}^{\mathrm{rain}}(P) \triangleq \mathrm{median}\big\{\mathrm{RAE}_{P,s}(p) : p \in P_{\mathrm{rain}},\ d_3(p) \in B_i\big\}\ .$$

If a patch contains no rainy pixels in a particular distance bin, this median is undefined and that patch does not contribute to the summary for that bin. Consequently, the number of contributing patches may differ between bins.

In Figure 3, these patch-level medians are compared across the full benchmark collection $\mathcal{P}$ of 100 patches. There, the displayed distance bins are

$$[0, 125]\ \mathrm{m},\ (125, 375]\ \mathrm{m},\ (375, 750]\ \mathrm{m},\ (750, 1500]\ \mathrm{m},\ (1500, 3125]\ \mathrm{m},\ (3125, 6000]\ \mathrm{m},\ (6000, \infty)\ \mathrm{m}.$$

In the box-and-whisker plot, the two sparsely populated tail bins $(6000, 9000]$ m and $(9000, \infty)$ are merged and displayed as a single $(6000, \infty)$ m bin. The count table below still reports these two tail bins separately.

For fixed solver s and distance bin B_i, let

$$\mu_{s,i}^{\mathrm{rain}} \triangleq \{m_{s,i}^{\mathrm{rain}}(P) : P \in \mathcal{P},\ m_{s,i}^{\mathrm{rain}}(P)\ \text{is defined}\}$$

denote the sequence of patch-level rainy-pixel median RAE values across patches. Each box-and-whisker then summarizes this sequence: the lower whisker tip is the minimum of $\mu_{s,i}^{\mathrm{rain}}$, the lower edge of the box is its 25th percentile, the horizontal line segment with the filled circle at its center marks the median, the upper edge of the box is its 75th percentile, and the upper whisker tip is its maximum.

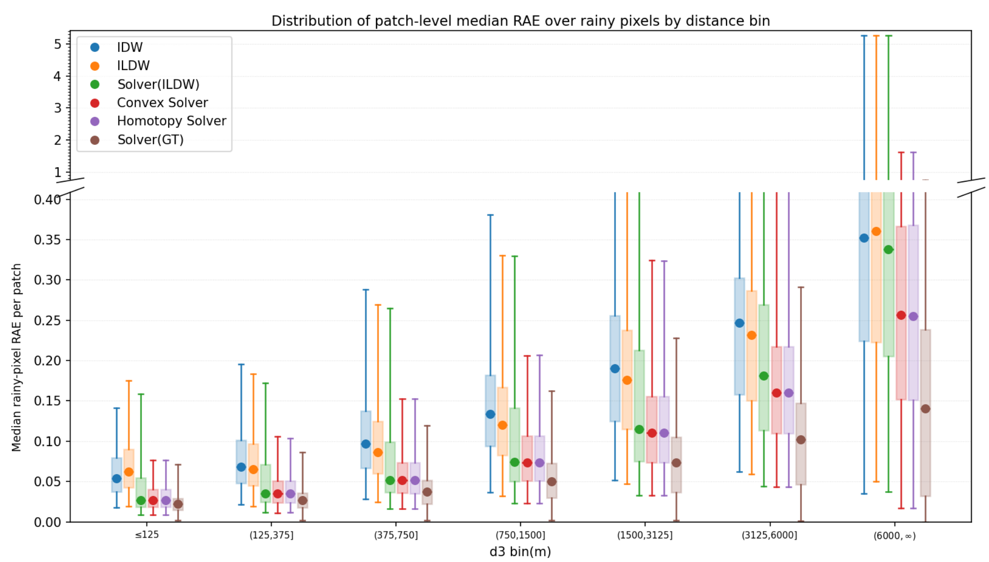


Figure 3: Rainy-pixel distance-profile box-and-whisker summary for distance to the third closest link. Each patch containing at least one rainy pixel in a given distance bin contributes one median rainy-pixel relative-absolute-error (RAE) value for that bin, and these patch-level medians are then summarized across patches with a box-and-whisker plot. Rainy pixels are those whose ground-truth rainfall is at least the configured wet threshold, set here to 0.6 mm/h. The corresponding rainy-pixel counts are reported separately in Table 6.

Distance to third closest link (m)	Rainy mean count (% of total)	SD
$[0, 125]$	4267.0 (1.79%)	960.4
$(125, 375]$	16737.7 (7.01%)	3803.2
$(375, 750]$	26809.9 (11.22%)	6707.0
$(750, 1500]$	51988.5 (21.76%)	14 468.0
$(1500, 3125]$	88839.4 (37.18%)	27 531.1
$(3125, 6000]$	45929.4 (19.22%)	15 679.4
$(6000, 9000]$	3835.8 (1.61%)	2725.7
$(9000, \infty)$	514.4 (0.22%)	966.9
Total	238922.1 (100%)	

Table 6: Rainy-pixel counts per distance bin for the same distance-to-third-closest-link partition used in Figure 3. For each patch, pixels are classified by the value of $d_3(p)$, and the table reports the mean and standard deviation of the rainy-pixel count in each bin across patches.

4.3.2 Non-Rainy Pixel Absolute Error by Link Distance

For the dry-pixel analysis, we focus on pixels whose ground-truth rainfall falls below the wet threshold. We therefore define the non-rainy pixel set by

$$P_{\text{nonrain}} = \{p \in P : R_P(p) < 0.6\}.$$

For these pixels, relative normalization by $R_P(p)$ is not appropriate because the denominator may be zero or very small. We therefore measure prediction quality by absolute error rather than relative absolute error:

$$\text{AE}_{P,s}(p) = |R_P(p) - \widehat{R}_{P,s}(p)|.$$

Using the same definition of the third-closest-link distance $d_3(p)$ introduced above, and the same distance bins B_i as in the rainy-pixel analysis, we compute for each patch P, solver s, and distance bin B_i

$$m_{s,i}^{\text{nonrain}}(P) \triangleq \text{median}\big\{\text{AE}_{P,s}(p) : p \in P_{\text{nonrain}},\ d_3(p) \in B_i\big\},$$

and, for fixed s and i, the corresponding box-and-whisker plot summarizes the sequence

$$\mu_{s,i}^{\text{nonrain}} \triangleq \{m_{s,i}^{\text{nonrain}}(P) : P \in \mathcal{P},\ m_{s,i}^{\text{nonrain}}(P) \text{ is defined}\}$$

of patch-level non-rainy-pixel median AE values across patches. As in the rainy-pixel analysis, patches without eligible pixels in a bin are omitted from that bin's summary. Thus the rainy and non-rainy figures use the same distance-bin construction, but they differ both in the pixel set being summarized and in the error metric being aggregated.

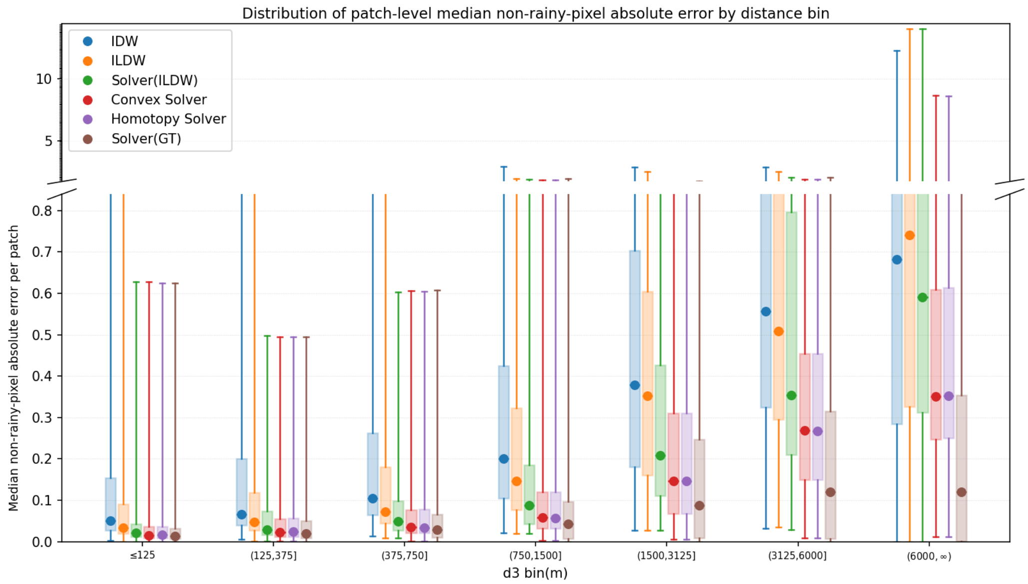


Figure 4: Non-rainy-pixel distance-profile box-and-whisker summary for distance to the third closest link. Each patch containing at least one non-rainy pixel in a given distance bin contributes one median non-rainy-pixel absolute error value for that bin, and these patch-level medians are then summarized across patches with a box-and-whisker plot. Non-rainy pixels are those whose ground-truth rainfall is below the configured wet threshold, set here to 0.6 mm/h. The corresponding non-rainy-pixel counts are reported separately in Table 7.

Distance to third closest link (m)	Non-rainy mean count (% of total)	SD
$[0, 125]$	563.6 (1.84%)	777.6
$(125, 375]$	2261.1 (7.40%)	3093.5
$(375, 750]$	3389.1 (11.09%)	4709.5
$(750, 1500]$	6446.5 (21.09%)	9257.7
$(1500, 3125]$	10776.2 (35.25%)	15 553.7
$(3125, 6000]$	6197.2 (20.27%)	8990.0
$(6000, 9000]$	849.9 (2.78%)	1289.7
$(9000, \infty)$	85.5 (0.28%)	250.9
Total	30569.2 (100%)	

Table 7: Non-rainy-pixel counts per distance bin for the same distance-to-third-closest-link partition used in Figure 4. For each patch, pixels are classified by the value of $d_3(p)$, and the table reports the mean and standard deviation of the non-rainy-pixel count in each bin across patches.

To make the distance-conditioned summaries easier to interpret geometrically, we show the distance-bin assignment for the largest evaluated patch. This illustrates how the third-closest-link distance partitions a real patch into near-link and low-coverage regions.

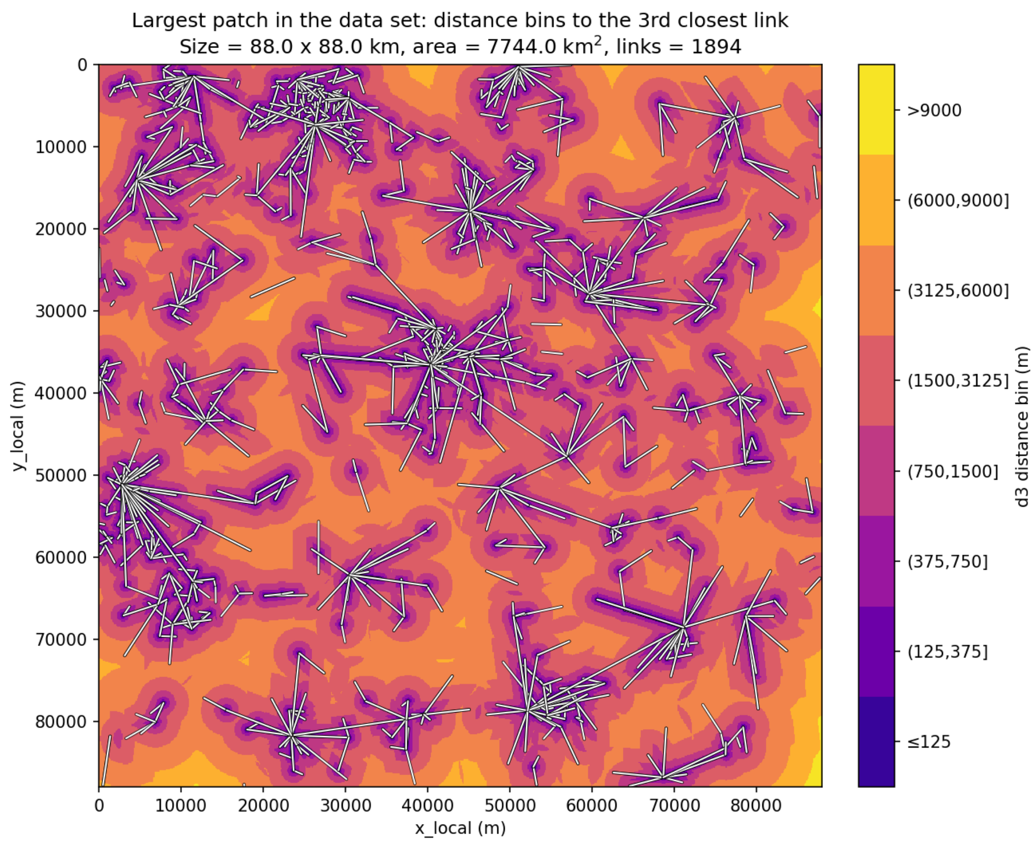


Figure 5: Distance-bin map for the largest evaluated patch in the data set. Each pixel is colored according to the bin induced by its distance to the third closest link, and the complete link geometry is overlaid in white.

4.4 Wet/Dry Confusion Analysis

Beyond pixel-level error magnitudes, we also summarize whether each method preserves the basic wet/dry structure of the rainfall field. Using the same 0.6 mm/h threshold as above, each pixel is classified as wet or dry in both the ground truth and the reconstruction. This gives patch-level true-positive, false-negative, false-positive, and true-negative fractions, summarized in Table 8.

False negatives warrant particular attention because they correspond to missed rain, while false positives create spurious wet areas. Wet/dry rainfall classification generally involves a trade-off between these errors, and their relative importance depends on the intended application (Polz et al., 2020). Because these rain-event patches contain approximately 89.5% wet pixels on average, normalizing all four entries by the total patch size makes the false-positive fractions shown in Table 8 appear relatively small. Conditional on ground-truth dry pixels, the mean of the patch-level conditional false-positive rates is approximately 59.8% for IDW and 40.0% for Convex Solver. Conditional on ground-truth wet pixels, the mean of the corresponding patch-level conditional false-negative rates remains small, at approximately 0.32% and 0.50%, respectively, meaning that both methods identify at least 99.5% of ground-truth wet pixels. Thus, Convex Solver remains close to IDW in wet-pixel detection, with a slightly higher false-negative rate, while substantially reducing false wet-area detection.

Solver	TP/$\|P\|$	FN/$\|P\|$	FP/$\|P\|$	TN/$\|P\|$
IDW	0.8926 ± 0.0126	0.0024 ± 0.0004	0.0396 ± 0.0039	0.0654 ± 0.0096
ILDW	0.8921 ± 0.0127	0.0029 ± 0.0004	0.0370 ± 0.0037	0.0680 ± 0.0097
Solver(ILDW)	0.8919 ± 0.0127	0.0031 ± 0.0004	0.0324 ± 0.0036	0.0726 ± 0.0098
Convex Solver	0.8912 ± 0.0127	0.0038 ± 0.0005	0.0228 ± 0.0022	0.0823 ± 0.0109
Homotopy Solver	0.8912 ± 0.0127	0.0038 ± 0.0005	0.0228 ± 0.0022	0.0822 ± 0.0109
Solver(GT)	0.8929 ± 0.0125	0.0021 ± 0.0003	0.0114 ± 0.0013	0.0936 ± 0.0123

Table 8: Patch-averaged wet/dry confusion matrix entries at a threshold of 0.6 mm/h, where a pixel is called wet if its rainfall is at least 0.6 mm/h and dry otherwise. For each patch, pixels are first assigned to the four confusion categories defined by the ground-truth and predicted wet/dry labels. The count in each category is then divided by the total number of pixels in that patch, producing a patch-level pixel fraction. The table entries report the mean of those patch-level fractions across the 100 evaluated patches, together with the standard error of the mean (SEM).

4.5 Reconstruction Runtime

Having compared reconstruction quality across the preceding analyses, we next assess the computational feasibility of Convex Solver for near-real-time applications. We timed each of the 100 benchmark patches using Convex Solver, with IDW and ILDW included for comparison. This focused experiment was not intended as an exhaustive runtime comparison across all solver variants; Solver(ILDW), Homotopy Solver, and the diagnostic Solver(GT) were therefore not timed. The patches and methods were processed sequentially on an Apple M2 MacBook Air with an 8-core CPU and 8 GB of memory. Each measurement is an end-to-end wall-clock time that includes input loading, initialization, reconstruction, and writing the reconstructed rainfall map and associated solver outputs. The resulting runtime summaries are reported in Table 9.

Solver	Mean ± SD (s)	Median (s)	Min–max (s)
IDW	7.53 ± 2.43	7.10	4.01–14.18
ILDW	18.69 ± 6.20	18.25	8.67–38.19
Convex Solver	76.03 ± 27.08	71.31	32.76–153.12

Table 9: End-to-end wall-clock reconstruction times across the 100 benchmark patches. The methods and patches were run sequentially on an Apple M2 MacBook Air with an 8-core CPU and 8 GB of memory.

Summary of findings. Across the benchmark, the optimized solvers have lower patch-level RMSE and higher Pearson correlation than IDW. Convex Solver and Homotopy Solver also give stronger rainfall-map agreement than Solver(ILDW) while maintaining strong attenuation consistency. Homotopy Solver offers no map-level improvement over Convex Solver, only a small gain in attenuation consistency. Solver(GT) has the strongest map-level summaries, whereas Convex Solver and Homotopy Solver achieve lower attenuation mismatch, confirming that the map-level and link-based measures capture different aspects of reconstruction quality.

Errors generally increase as nearby-link coverage becomes weaker, but the optimized solvers retain lower median errors than IDW in the low-coverage bins. Convex Solver reduces false wet-area detection relative to IDW at the cost of a small increase in missed-rain pixels. Its 76.03-second mean runtime suggests potential for near-real-time use at the benchmark scale.

Chapter 5

Conclusion and Outlook

This thesis compared interpolation and optimization-based reconstruction strategies on a 100-patch benchmark constructed from the OPERA-based EURADCLIM dataset and simulated microwave-link attenuations. Across the reported experiments, the optimized solvers achieved lower attenuation mismatch than IDW when all final reconstructions were evaluated using the attenuation model with each link's original frequency and polarization. Convex Solver and Homotopy Solver also had lower map-level RMSE, lower mean signed bias, and higher Pearson correlation summaries than both IDW and the ILDW-initialized nonlinear solver. Solver(GT) had the strongest reported map-level summaries, but it is initialized from the ground truth and therefore is not a deployable method. At the same time, the comparison between Solver(GT) and the convex solvers shows that the rainfall field closest to the radar-derived ground truth is not always the field that best matches the link observations under the chosen forward model. ILDW remained a useful link-aware IDW modification: it preserves the simplicity of IDW-style interpolation while making better use of the fact that each CML observation belongs to a full link path rather than only to a midpoint.

An important next step is to repeat the comparison of reconstruction methods using measurements from an operational CML network. In the current benchmark, the observed attenuations are generated from the radar-derived ground-truth rainfall fields using the attenuation model with each link's original frequency and polarization, producing clean and model-consistent observations. In a real-time application, the attenuation-mismatch term would instead be driven by rain-induced attenuations estimated from current CML signal measurements, while independent radar or rain-gauge observations would be withheld from the reconstruction and used only afterward as reference data for evaluating the resulting rainfall estimates. Moreover, operational CMLs record attenuation as rainfall varies at shorter time scales. Because the rain-rate–attenuation relation is nonlinear, attenuation computed from an hourly mean rain rate is not generally equal to the mean attenuation produced by the varying rain rates within that hour. Such an experiment is needed to determine whether the improvements over IDW persist in the presence of measurement noise, changing link availability, wet-antenna attenuation, baseline drift, mismatch between the forward model and real signal behavior, and temporal aggregation effects.

The runtime results in Table 9 provide an initial indication of the computational feasibility of such an application. On the benchmark hardware, Convex Solver reconstructed a patch in an average of 76.03 seconds (median 71.31 seconds; range 32.76–153.12 seconds). This suggests that the solver runtime may be compatible with near-real-time reconstruction at the benchmark scale, although operational latency would also include collecting, delivering, and preprocessing the CML measurements.

Beyond this operational validation, one direction is to move from single-time reconstruction to longer event sequences over the same region and fixed link network. Even without changing the objective, this would show whether the method behaves stably over time or produces frame-to-frame artifacts when applied independently to consecutive observations. A second step would be to incorporate temporal structure directly. In meteorology, *optical flow* refers to methods that estimate an apparent motion field between consecutive images (Ayzel et al., 2019). For CML-based nowcasting, each set of link measurements at a discrete time would first be reconstructed into a rainfall map, and a sequence of these reconstructed maps would be used to estimate the motion field. Advecting the latest reconstructed map according to this field would provide a short-term forecast. Once the next set of CML measurements becomes available, the advected map could serve as a temporal prior: the reconstruction would be encouraged to remain close to the forecast, while the attenuation-mismatch term would allow it to depart where the new observations indicate that the forecast is inaccurate.

A further possibility would be to adapt the present reconstruction framework to radar–CML data fusion, building on existing variational fusion work (Bianchi et al., 2013). Radar observations could enter through a radar-mismatch term, combining radar's spatial coverage with the path-integrated, near-ground information provided by CMLs. This adaptation cannot be evaluated independently with the current benchmark because its EURADCLIM maps already serve as ground truth; an operational test would require real radar and CML measurements, verification data withheld from the solver, and careful calibration of the relative observation weights.

Another possible direction is to compare the deterministic solvers studied here with learning-based inverse approaches. Recent work, for example, has studied CML rain-field reconstruction using a Bayesian inverse formulation with learned generative priors (Moufad et al., 2026). Such methods could provide a useful point of comparison, but they would require a separate evaluation of training data, prior assumptions, and generalization beyond the benchmark setting used here.

Together, these directions would test whether the improvements observed under controlled, model-consistent conditions carry over to operational measurements and more realistic reconstruction settings.

Appendix A

Supplementary Analyses

A.1 Largest-Patch Link-Network Summary

This appendix summarizes the link-network properties of the largest evaluated patch shown in Figure 5. Selected by physical area, this is the 88 km by 88 km patch from January 29, 2023 at 19:00, with an area of 7744 km^2 and 1,894 retained links. The distance-bin map shows which pixels are close to or far from nearby links, but it does not by itself describe the link network that produced that geometry.

Here, link length refers to the path length of a link, carrier frequency is the microwave-link frequency in GHz, and endpoint degree is the number of links incident to a given endpoint node.

Link length matters because longer links average rainfall over longer paths and can smooth local rainfall variation. Carrier frequency matters because rain-induced attenuation depends on frequency. Endpoint degree gives a compact view of whether the network is mostly chain-like or whether many links meet at shared nodes.

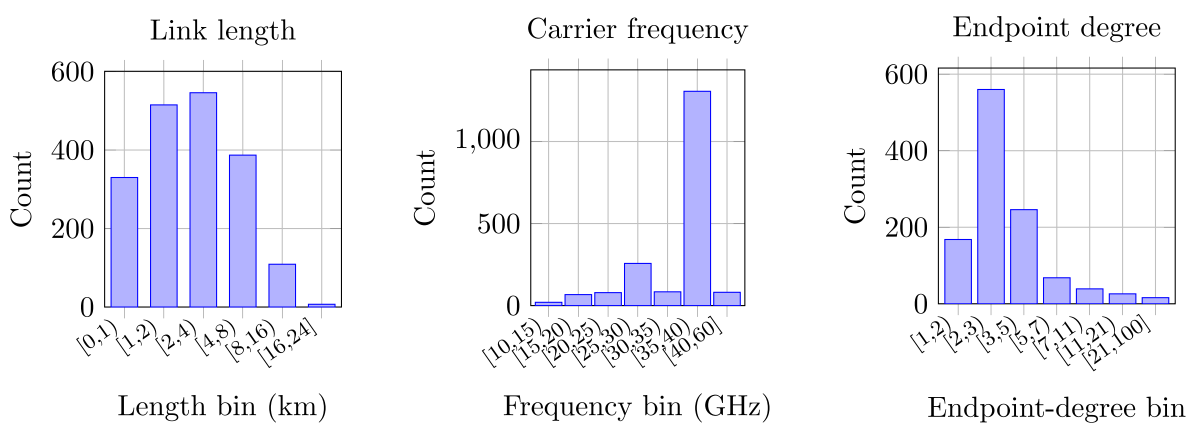


Figure 6: Histogram summaries for the largest patch. The left panel shows link path lengths, the middle panel shows carrier frequencies, and the right panel shows endpoint degrees, where endpoint degree is the number of links incident to a given endpoint node. These distributions provide context for the link geometry underlying the distance-bin map in the main text.

Of the 1,894 links, 73.4% are shorter than 4 km and 93.9% are shorter than 8 km; 69.0% operate between 35 and 40 GHz. Among the 1,123 distinct endpoints, 86.7% have degree below five. Thus, the network is dominated by short, 35–40 GHz links, and most endpoints have low degree.

A.2 Case Studies

Across the patches inspected manually, two recurring qualitative patterns emerged. In the first, Solver(GT) remains close to the ground-truth rainfall field while Convex Solver differs visibly. In the second, Solver(GT) and Convex Solver produce similar reconstructions, both visibly different from the ground-truth rainfall field. The two cases below illustrate these patterns.

The first example illustrates the first pattern. In the Black Sea coast, Romania case study, Solver(GT) closely resembles the ground-truth rainfall field, while Convex Solver gives a visibly different reconstruction. The table compares their final fields using J_{atten}:

Solver	J_{atten}	RMSE (mm/h)
Solver(GT)	1.81×10^{-5}	0.102
Convex Solver	4.06×10^{-6}	0.702

For a complementary within-solver view, Solver(GT)'s own weighted objective evolves as follows:

Solver(GT) state	Solver(GT) weighted objective	RMSE (mm/h)
Ground-truth initialization	2.832	0.000
Final reconstruction	1.485	0.102

Convex Solver has the lower J_{atten}, but Solver(GT) is substantially closer to the radar-derived ground-truth field. Thus, stronger consistency with the link observations does not necessarily imply lower pixelwise reconstruction error. Moreover, Solver(GT)'s increasing RMSE as its own objective decreases shows that this distinction also arises within a single formulation.

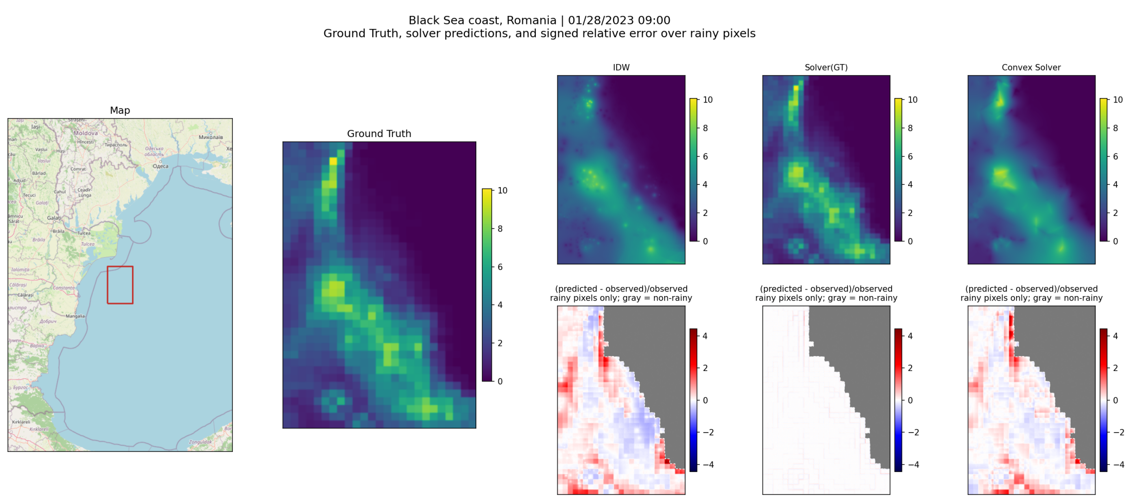
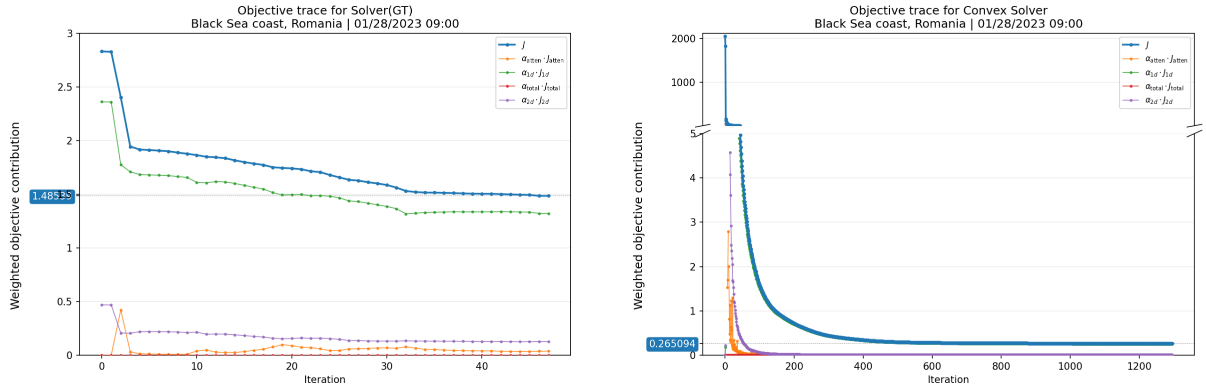


Figure 7: Black Sea coast, Romania case study observed on January 28, 2023 at 09:00. The panel shows the ground truth, IDW, Solver(GT), and Convex Solver.



(a) Solver(GT) (b) Convex Solver

Figure 8: Solver-specific objective traces for the Black Sea coast, Romania case study. Values can be compared over iterations within each panel, but not directly between panels because the solvers use different link models and objective normalizations.

The traces show that most of the objective reduction occurs early. For Convex Solver, the objective continues to decrease after several hundred iterations, but the visible change after the first few hundred iterations is small and is driven mainly by the first-difference penalty J_{1d}. Solver(GT) stops after 47 iterations when the ftol criterion is met, following the initial objective reduction.

The second example illustrates the second pattern. In the Northern Norway case study, Solver(GT) and Convex Solver closely resemble each other and have similar J_{atten} and RMSE values:

Solver	J_{atten}	RMSE (mm/h)
Solver(GT)	1.24×10^{-6}	0.552
Convex Solver	2.50×10^{-6}	0.565

Solver(GT)'s own trajectory is:

Solver(GT) state	Solver(GT) weighted objective	RMSE (mm/h)
Ground-truth initialization	3.142	0.000
Final reconstruction	0.289	0.552

Here Solver(GT) is slightly better than Convex Solver on both common evaluation measures, although the differences are small. As in the first case, however, Solver(GT) moves away from its exact ground-truth initialization while lowering its own weighted objective.

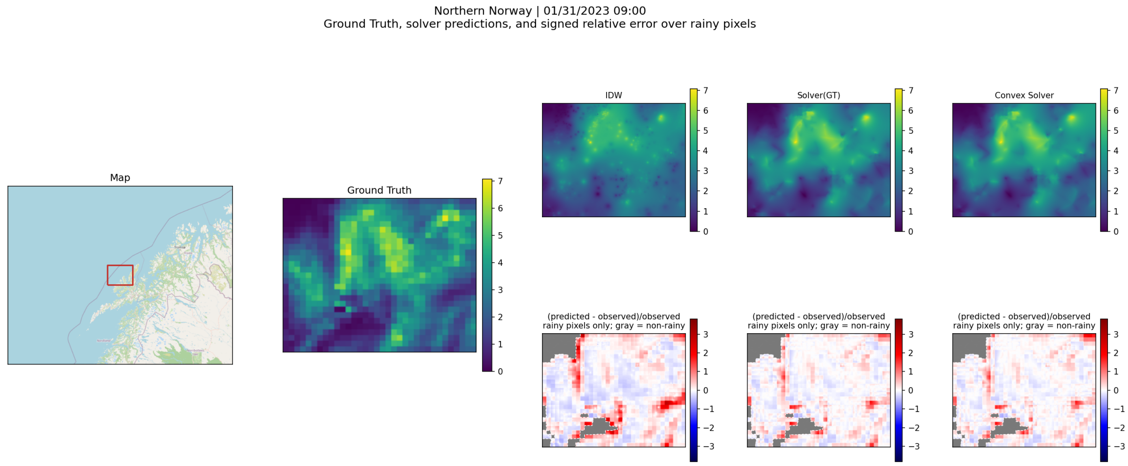
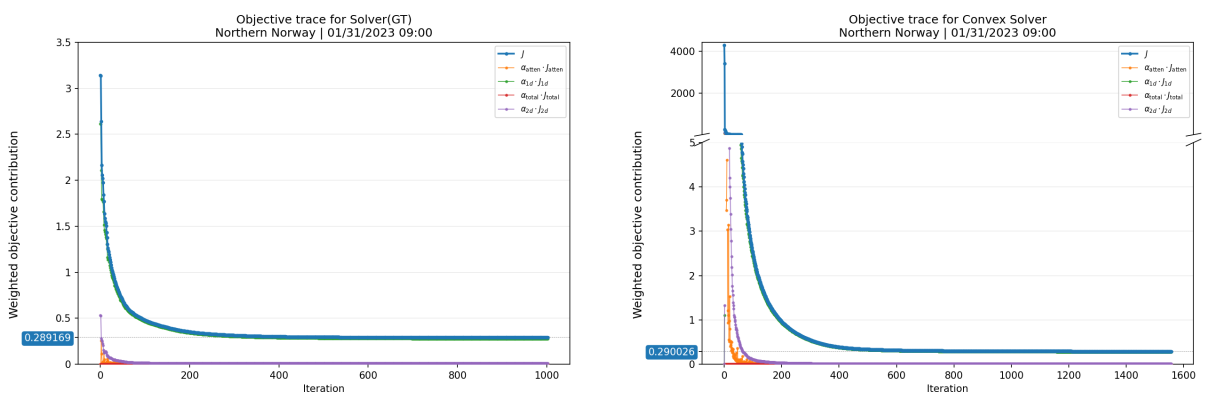


Figure 9: Northern Norway case study observed on January 31, 2023 at 09:00. The panel shows the ground truth, IDW, Solver(GT), and Convex Solver.

In this case, the error heatmaps show smaller areas of positive error for Solver(GT) and Convex Solver than for IDW.



(a) Solver(GT)

(b) Convex Solver

Figure 10: Solver-specific objective traces for the Northern Norway case study. As above, objective values should be interpreted within each solver rather than compared directly across panels.

As in the Black Sea case, the first few hundred Convex Solver iterations account for most of the visible objective reduction, with the remaining gradual decrease driven mainly by the first-difference penalty J_{1d}.

Taken together, these appendix case studies support the main-text interpretation that agreement with the link observations and pixelwise agreement with the radar-derived ground truth capture different aspects of reconstruction quality. In the first case, Convex Solver has lower J_{atten} but substantially higher RMSE than Solver(GT); in the second, the two methods are close on both measures and produce similar reconstructions. In both cases, Solver(GT) reduces its own objective while increasing its pixelwise error from zero.

Statement on the Usage of Generative Digital Assistants

For this thesis, I used OpenAI ChatGPT and OpenAI Codex as generative digital assistants. Their use included support for programming and debugging Python and LaTeX code, rendering and checking report artifacts, converting numerical summaries into LaTeX tables, comparing generated figures and output files, and revising explanatory prose for clarity, structure, and readability.

The tools were used as assistants rather than as independent scientific sources. Mathematical formulations, empirical results, numerical values, figures, citations, and final interpretations were checked against the project code, generated artifacts, local files, and the cited literature. I reviewed and edited the thesis text and take full responsibility for the submitted content, including the accuracy and completeness of its citations and references.

I am aware of the potential risks of using generative digital assistants, including incorrect suggestions, hallucinated claims, and over-polished wording that may obscure uncertainty. For this reason, I used them with caution and critical review.

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